

Problem

Submissions

Leaderboard

Discussions

Given a text `txt[0..n-1]` and a pattern `pat[0..m-1]`, write a function `search(char pat[], char txt[])` that prints all occurrences of `pat[]` in `txt[]`. You may assume that $n > m$.

Input Format

`txt[]="THIS IS A TEST TEXT"` `pat[]="TEST"`

Constraints

TIME COMPLEXITY : $O(N)$

Output Format

Pattern found at index 10

Sample Input 0

AABAACAADAABAABA
AABA

Sample Output 0

0
9
12

Sample Input 1

AABAACAADAABAAABAA
AABA

Sample Output 1

0
9
13

Change Theme

Language

Python 3



```
1 def search(pat, txt):
2     s = pat + "$" + txt
3     n = len(s)
4     m = len(pat)
5     z = [0] * n
6     l = r = 0
7     for i in range(1, n):
8         if i <= r:
9             z[i] = min(r - i + 1, z[i - l])
10
11         while i + z[i] < n and s[z[i]] == s[i + z[i]]:
12             z[i] += 1
13
14         if i + z[i] - 1 > r:
15             l = i
16             r = i + z[i] - 1
17
18         if z[i] == m:
19             print(i - m - 1)
20
21 txt = input()
22 pat = input()
23 search(pat, txt)
```

Line: 23 Col: 5

 Upload Code as File

Run Code

Submit Code

☒ Test against custom input

```
aaabacdabcbabda
ab
```

Congratulations

You solved this challenge. Would you like to challenge your friends?

☒ Test case 0

Compiler Message

☒ Test case 1

Success



https://www.hackerrank.com/contests/2026-competitive-coding-lab1/challenges/pattern-searching-5/problem?isFullScreen=true

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